

Cross-reference & verification engine — BUILT v0.79.0

One provenance-tracked part record. FLIS-validated name · vehicles it fits · interchange · manufacturer · breakdown image.

❏ X1 · FLIS repair ✓

- NSN→official name (ref_nsn). PREFERRED, OCR kept visible,
- conflict FLAGGED. 'HOSE NONMET' → HOSE, NONMETALLIC.
- + PN+CAGEC→NSN recovery from PUB LOG (pn_nsn.json).

Verified, conf 0.9.

❏ X2 · Vehicle assignment ✓

- NSN→vehicles via nsn_platforms (45,068), noise-filtered.
- M915 / HMMWV bound to the part + its breakdown.
- Many-to-many.

The 'match to vehicles' ask.

❏ X3 · Asset cross-link ✓

- One record links: breakdown image, schematic, procedure,
- look-alike, tags, manufacturer (CAGEC→company, cage.json),
- interchangeable + superseded.

Everything, by NSN/PN.

❏ X4 · Online enrich ✓ (off by default)

- Public-data-only, CACHED (xref_online.json), ITAR/EAR-aware.
- Gated by VIEWER_XREF_ONLINE + your endpoint; app only READS
- the cache, never fetches while serving. docs/XREF-ONLINE-SETUP.

Opt-in, never runtime.

❏ X5 · Provenance & review ✓

- Every field tags its SOURCE; record carries CONFIDENCE.
- FLIS-vs-OCR conflicts flagged in the part card (⚠ OCR read).
- Residual → ❏ Part# review queue (bulk fix, your fixes win).

Nothing silently wrong.

VERIFIED + how to run

part_record('12420572-010') on a fixture: PN→NSN, FLIS name preferred (OCR kept + conflict flagged), vehicles M915/HMMWV (junk filtered), interchangeable NSN, confidence 0.9, full provenance; cage.json/xref_online.json reads add manufacturer + colloquial. Run order on your PC: BUILD-RPSTL.bat → BUILD-XREF.bat (PUB LOG) → search a part number. X4 stays off unless you set VIEWER_XREF_ONLINE. Routes in RUN-ALL-TESTS.bat; index never written (R1/R6).